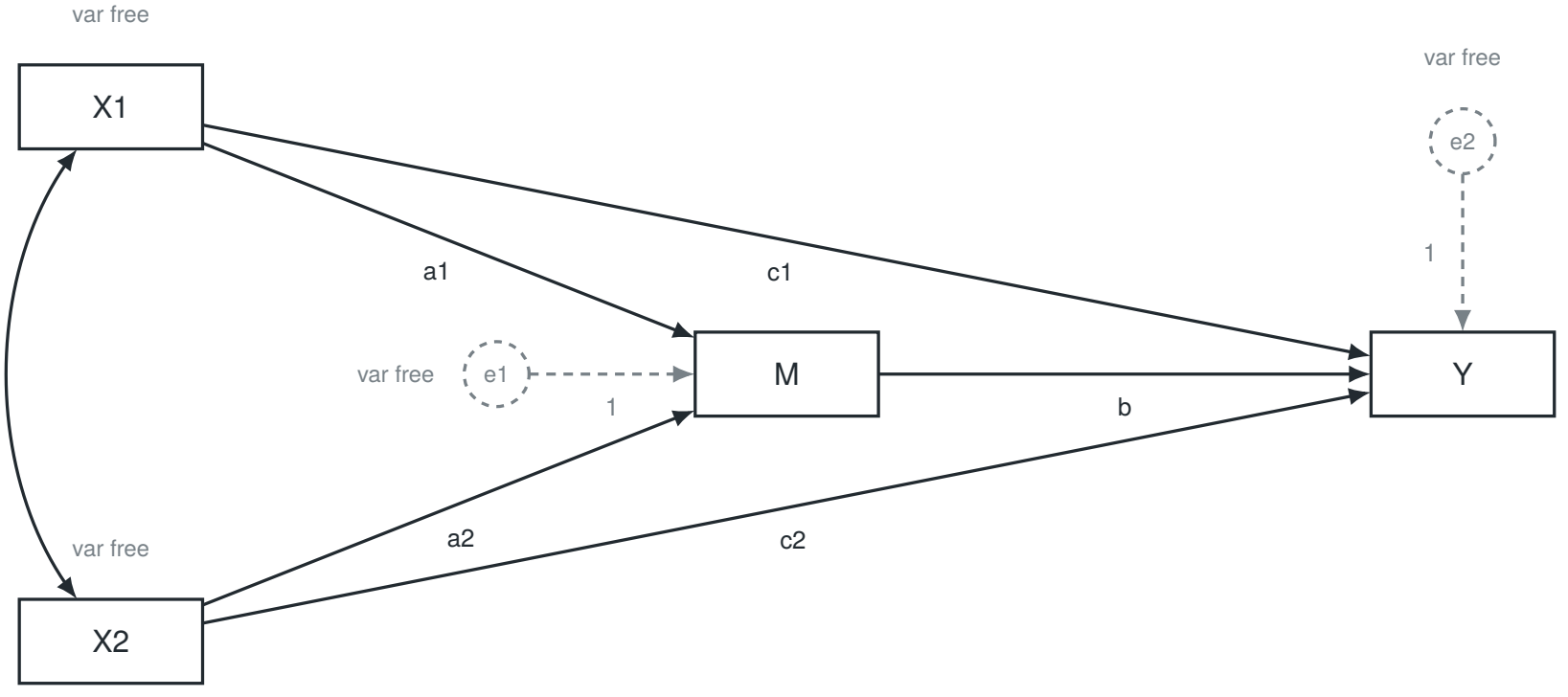Observed-only path model